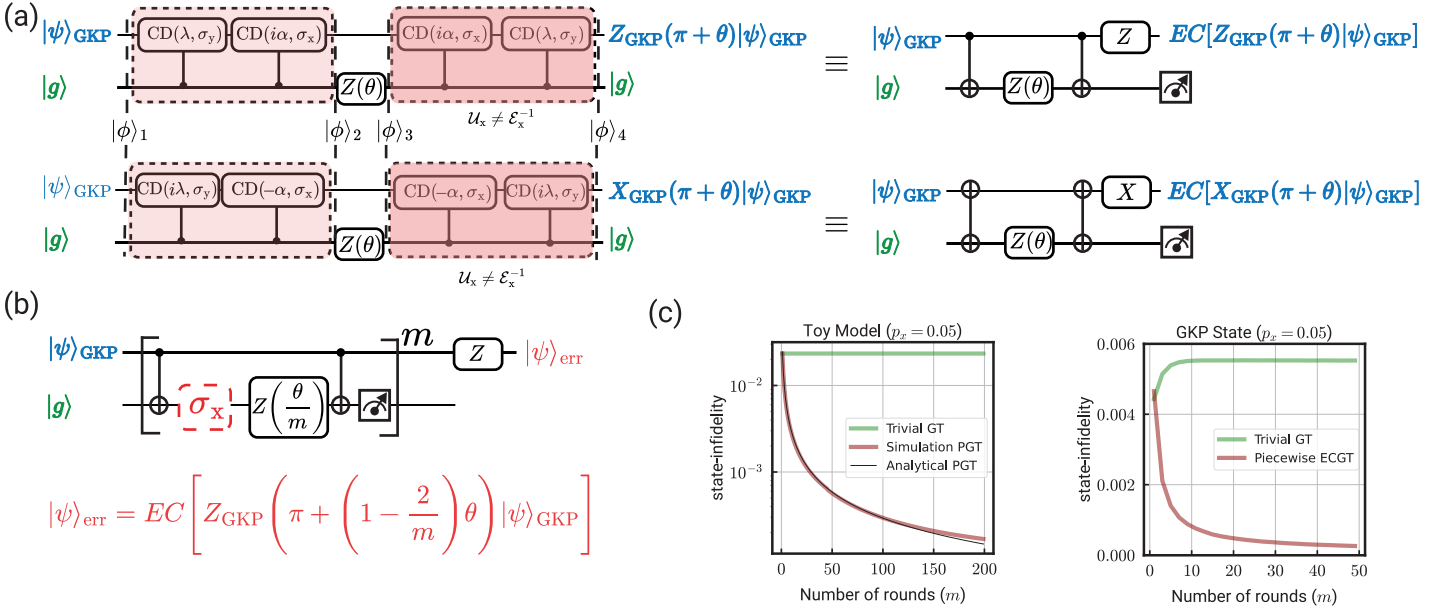

(a)
$|\psi\rangle_{\mathrm{GKP}}$
$|g\rangle$
$\mathrm{CD}(\lambda,\sigma_y)$
$\mathrm{CD}(i\alpha,\sigma_x)$
$Z(\theta)$
$\mathrm{CD}(i\alpha,\sigma_x)$
$\mathrm{CD}(\lambda,\sigma_y)$
$Z_{\mathrm{GKP}}(\pi+\theta)|\psi\rangle_{\mathrm{GKP}}$
$|\phi\rangle_1$
$|\phi\rangle_2$
$|\phi\rangle_3$
$|\phi\rangle_4$
$\mathcal{U}_x \neq \mathcal{E}_x^{-1}$
$\equiv$
Z
$EC[Z_{\mathrm{GKP}}(\pi+\theta)|\psi\rangle_{\mathrm{GKP}}]$
$\mathrm{CD}(i\lambda,\sigma_y)$
$\mathrm{CD}(-\alpha,\sigma_x)$
$\mathrm{CD}(-\alpha,\sigma_x)$
$\mathrm{CD}(i\lambda,\sigma_y)$
$X_{\mathrm{GKP}}(\pi+\theta)|\psi\rangle_{\mathrm{GKP}}$
X
$EC[X_{\mathrm{GKP}}(\pi+\theta)|\psi\rangle_{\mathrm{GKP}}]$
(b)
m
σ_x
$Z\left(\frac{\theta}{m}\right)$
$|\psi\rangle_{\mathrm{err}}$
$|\psi\rangle_{\mathrm{err}} = EC\left[Z_{\mathrm{GKP}}\left(\pi+\left(1-\frac{2}{m}\right)\theta\right)|\psi\rangle_{\mathrm{GKP}}\right]$
(c)
Toy Model ($p_x = 0.05$)
state-infidelity
10^{-2}
10^{-3}
Trivial GT
Simulation PGT
Analytical PGT
0
50
100
150
200
Number of rounds (m)
GKP State ($p_x = 0.05$)
state-infidelity
0.006
0.004
0.002
0.000
Trivial GT
Piecewise ECGT
0
10
20
30
40
50
Number of rounds (m)